Final Exam 2026: Part 2

63, 14

June 25, 2026

1 Introduction

This report studies minimum-variance portfolio allocation using high-frequency intraday data for the $N = 30$ constituents of the Dow Jones Industrial Average (DJIA) over the period 2008–2025. The minimum-variance portfolio is the constrained allocation

$$\omega_{MV}(\Sigma) = \arg\min_{\iota'\omega=1} \omega'\Sigma\,\omega = \frac{\Sigma^{-1}\iota}{\iota'\Sigma^{-1}\iota},$$

whose only input is the covariance matrix Σ. We compare two estimators of Σ: a **high-frequency (HF)** estimator built from intraday realized covariances (Andersen and Bollerslev, 1998; Barndorff-Nielsen et al., 2011), and a **Ledoit-Wolf (LW)** shrinkage estimator built from daily close-to-close returns (Ledoit and Wolf, 2003). The central question is whether the additional intraday information in the HF estimator translates into a measurably better out-of-sample portfolio.

2 Exercise 1: Volatility Signature Plot and Sampling Choice

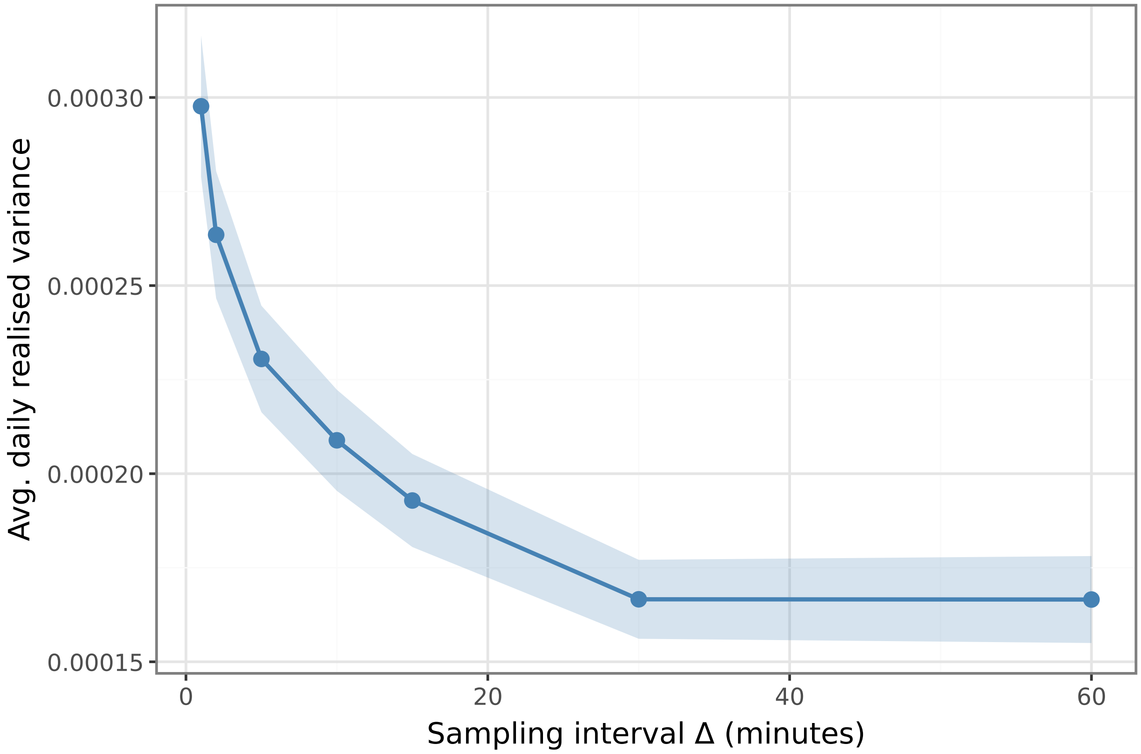

Figure 1: Volatility signature plot for AAPL. Average daily RV ± 2 SE. Bias at $\Delta = 1$ min reflects microstructure noise; stabilises at $\Delta = 30$ min.

The signature plot (Figure 1) shows the characteristic upward bias of the **realized variance (RV)** estimator at $\Delta = 1$ minute, where microstructure noise inflates squared log-returns above the true integrated variance ($IV \neq RV$) (Andersen and Bollerslev, 1998). Under a simple i.i.d. noise model, the signal-to-noise ratio at sampling interval Δ_n is

$$\pi = \frac{2a^2}{2a^2 + \sigma^2 \Delta_n},$$

where a^2 is the per-interval efficient-price variance and σ^2 is the noise variance; as Δ_n increases $\pi \to 1$, so coarser sampling reduces the noise contribution. The curve continues to decline noticeably up to $\Delta \approx 20$ minutes before flattening at $\Delta = 30$ minutes, motivating our sampling choice throughout; missing minutes are filled by **previous-tick interpolation**. We adopt 30-minute sparse sampling over noise-robust alternatives (TSRV, realized kernels) because the bias is already negligible at this frequency, while we retain useful intraday variation relative to daily returns.

3 Exercise 2: Daily Realized Covariance

The **realized covariance (RC)** estimator $RC_d = \sum_{k=1}^{K} r_k r_k'$ sums outer products of synchronised 30-minute log-return vectors (Andersen and Bollerslev, 1998; Barndorff-Nielsen et al., 2011).

Intraday gaps are filled by previous-tick interpolation; any remaining missing returns are set to zero, a conservative choice that avoids inflating RC_d.

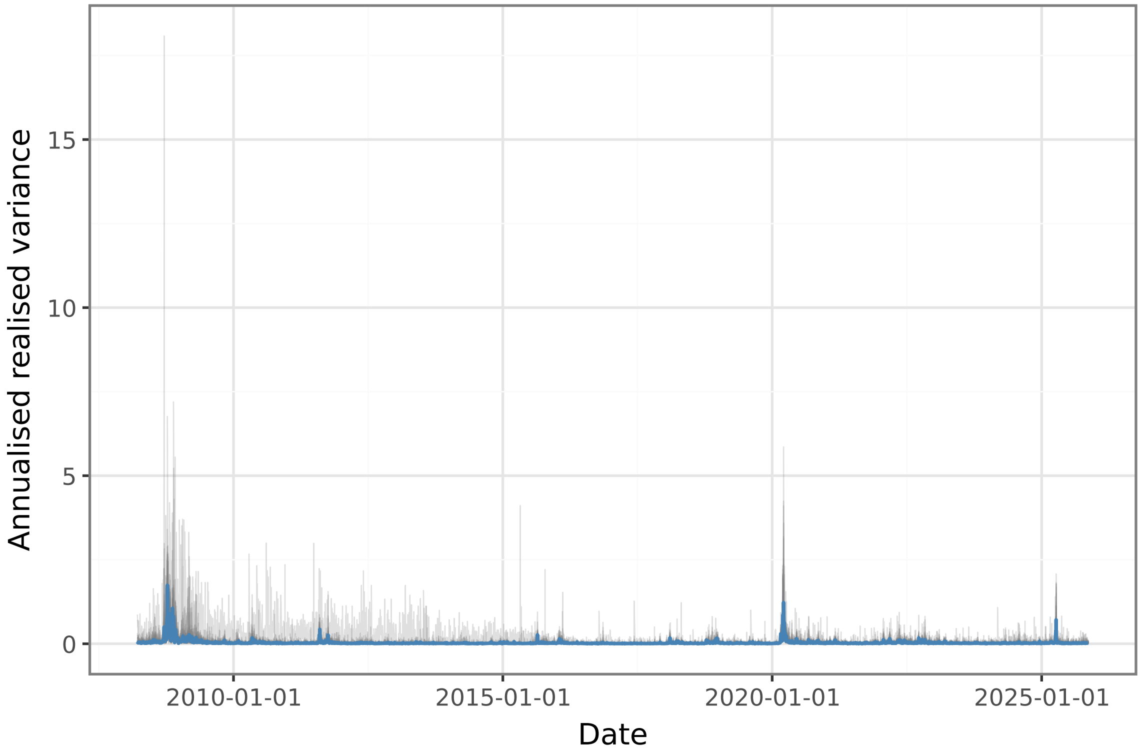

Figure 2: Annualised daily realised variance for all 30 DJIA stocks (grey), cross-sectional median (blue) and IQR (band). Spikes: 2008–09, March 2020, 2022.

The diagonal of RC_d (Figure 2) displays volatility clustering and persistent cross-sectional heterogeneity. The three largest spikes correspond to the 2008–09 crisis, the March 2020 COVID shock, and the 2022 rate-hike episode. At one-minute sampling, off-diagonal elements would be biased toward zero by the **Epps effect** (Barndorff-Nielsen et al., 2011); 30-minute sampling largely eliminates this.

4 Exercise 3: Covariance Estimators

The evaluation window runs from January 17, 2017 to November 14, 2025. **(a) HF random walk:** $\hat{\Sigma}_t^{HF} = RC_{t-1}$ uses yesterday's realized covariance as today's forecast. This implicitly assumes that the covariance matrix follows a **random walk**: $\Sigma_t = \Sigma_{t-1} + \varepsilon_t$, so that the best one-step-ahead predictor is the most recent observation. Equivalently, it treats the covariance as constant within each day and allows it to change arbitrarily between days, with no mean reversion or smoothing. **(b) LW on daily returns:** $W = 252$ days gives $T/N \approx 8.4$, ensuring a full-rank sample covariance; Ledoit and Wolf (2003) shrinkage toward a scaled identity stabilises the precision matrix for portfolio inversion.

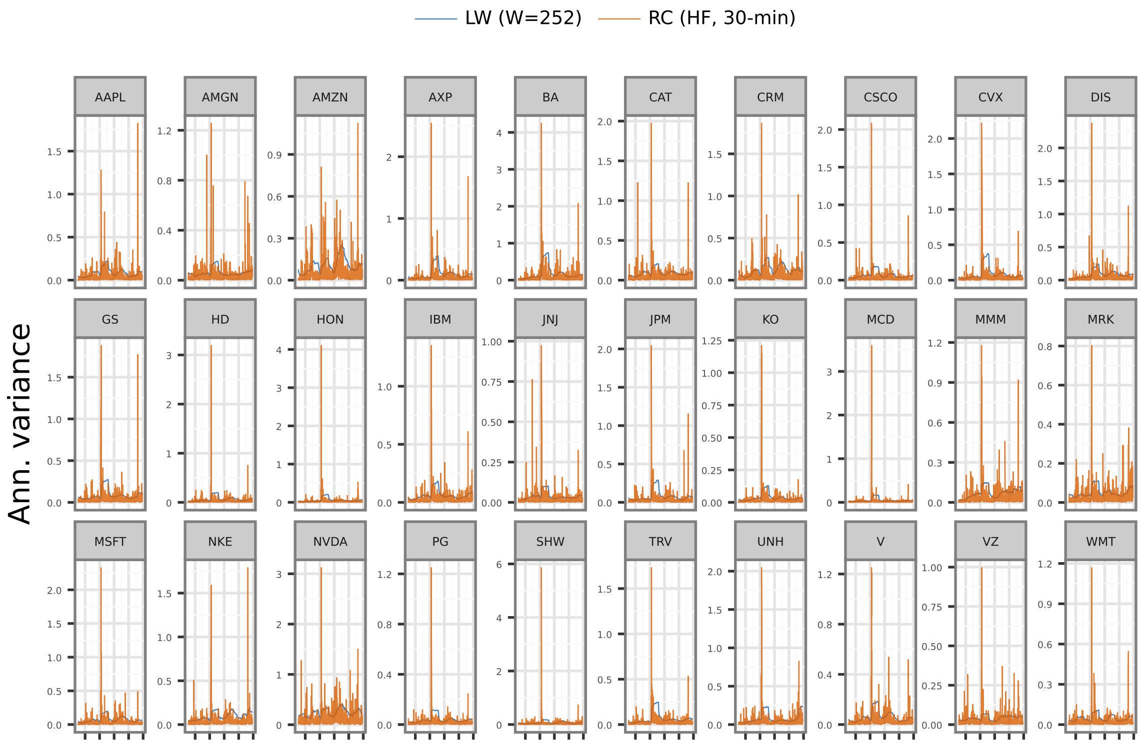

Figure 3: Per-stock annualised variance for all 30 DJIA stocks. LW (orange, rolling 252-day shrinkage) is consistently smoother; RC (blue, 30-min) reacts immediately to volatility bursts.

Figure 3 confirms the smoothing property of LW: the orange line is consistently flatter than the blue (RC) line, which spikes sharply during stress events. The free y-axis reveals cross-sectional heterogeneity, high-volatility stocks (e.g., NVDA, BA) spike far more than defensive names (e.g., KO, WMT).

5 Exercise 4: Minimum-Variance Portfolio Weights

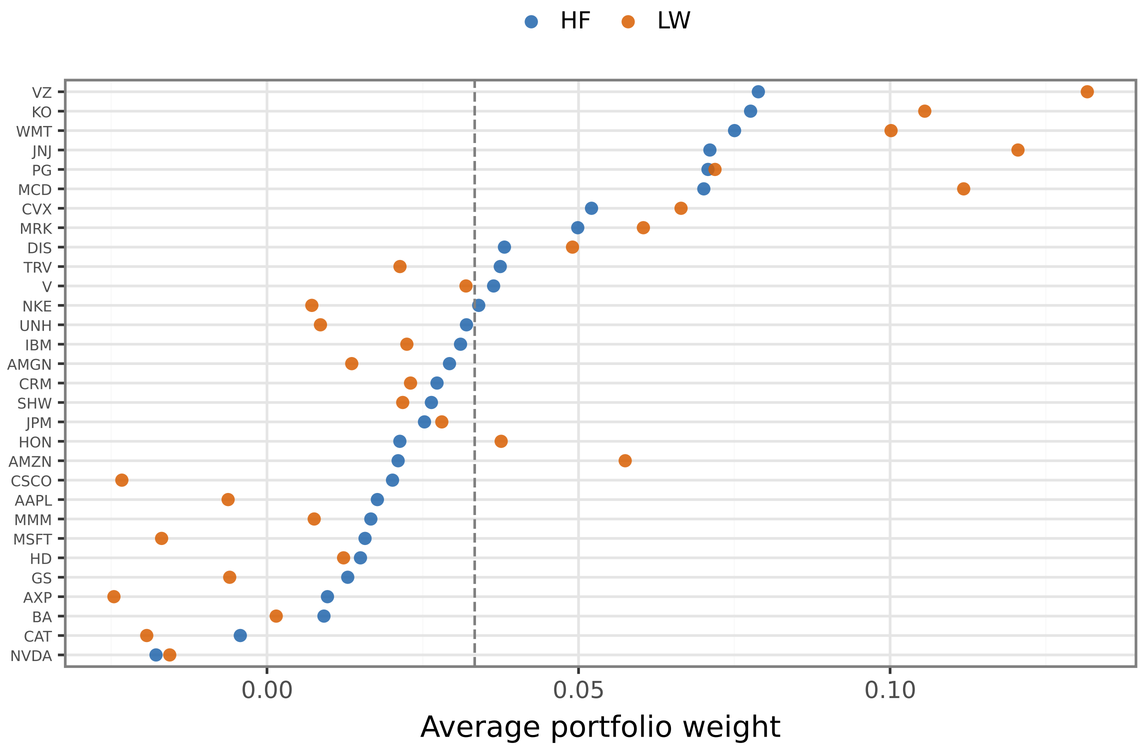

Figure 4: Time-averaged MV portfolio weights for 30 DJIA stocks, ordered by HF weight. Dashed line: equal weight (1/30).

The HF strategy assigns its largest average weight to **VZ** (0.079) and its most negative to **NVDA** (-0.018). The LW strategy is most positive on **VZ** (0.132) and most negative on **AXP** (-0.025). Average daily turnover is **3.422** (HF) and **0.071** (LW), with the HF strategy showing higher turnover.

Both strategies hold extreme positions, a known consequence of unconstrained MV optimisation (Ledoit and Wolf, 2003), where small estimation errors cause large weight oscillations. The notably high HF turnover reflects the fact that each day's RC_d is estimated from only \$ \$13 intraday returns, making it noisy and causing weights to shift substantially from one day to the next.

6 Exercise 5: Out-of-Sample Portfolio Evaluation

Table 1: Out-of-sample performance (second half of sample). Ann. volatility = s.d. $\times\sqrt{252}$; Sharpe ratio = ann. mean / ann. s.d. (risk-free rate = 0, no transaction costs).

Out-of-sample portfolio performance		
Strategy	Ann. Volatility	Ann. Sharpe
HF (RC, t-1)	0.191	0.572
LW (W = 252)	0.141	0.674
Equal-weighted (1/N)	0.179	0.763

Equal-weighted achieves the highest annualised Sharpe ratio (0.763); **HF (RC, t-1)** produces the lowest (0.572). The HF estimator does not outperform LW on Sharpe ratio (0.572 vs. 0.674).

The random-walk estimator uses only a single day per forecast; in the post-2016 window, LW's smoothing advantage dominates, producing more stable weights and a higher Sharpe ratio. HF covariance information adds most value when $N \gg 30$ and the true covariance shifts rapidly (Ledoit and Wolf, 2004); with only 30 DJIA stocks, LW already captures the dominant structure.

More broadly, these results illustrate a recurring finding in the portfolio choice literature: statistical sophistication in covariance or return estimation does not automatically translate into out-of-sample economic gains. The equal-weighted benchmark achieves the highest Sharpe ratio here, consistent with the observation that estimation error in large covariance matrices can amplify, rather than reduce, portfolio risk. Estimators that exploit additional information, such as intraday returns, shrinkage targets, or factor structures, tend to deliver meaningful improvements when the signal-to-noise ratio is high: either the universe is large enough that noise in the sample covariance is severe, or the true covariance structure is time-varying rapidly enough that high-frequency updates are valuable. With $N = 30$ liquid, well-studied blue-chip stocks over a relatively calm post-2016 sample, neither condition is strongly met, so the naive $1/N$ rule proves difficult to beat (Ledoit and Wolf, 2004).

7 Conclusion

This report compared two covariance estimators for minimum-variance portfolio allocation across 30 DJIA stocks. The HF random-walk estimator exploits intraday information but produces highly unstable weights and the lowest out-of-sample Sharpe ratio, while the LW estimator, despite relying only on daily returns, delivers lower portfolio volatility and a higher Sharpe ratio through its smoothing of the covariance matrix. Neither optimised strategy outperforms the naive equal-weighted benchmark, underscoring that with a modest universe of liquid stocks, statistical sophistication in covariance estimation does not automatically translate into economic gains. Future work could explore noise-robust HF estimators such as realized kernels, longer evaluation windows, or larger universes where high-frequency covariance information is more likely to confer a competitive edge.

8 References

Andersen, T. G. and Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4):885–905.

Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., and Shephard, N. (2011). Multivariate realised kernels: Consistent positive semi-definite estimators of the covariation of equity prices with noise and non-synchronous trading. *Journal of Econometrics*, 162(2):149–169.

Ledoit, O. and Wolf, M. (2003). Improved estimation of the covariance matrix of stock returns with an application to portfolio selection. *Journal of Empirical Finance*, 10(5):603–621.

Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411.